Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years shipping AI products end to end in **TypeScript/Node.js**, **Python**, and **Next.js**/React, turning stakeholder problems into LLM workflows, routing, and analytics with measurable business impact.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Advanced NLP, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*
- Shipped a multi-tenant **AI education platform** on Azure SQL and Blob Storage for 500+ users, designing row-level-secure schemas and tuning queries that cut dashboard latency from 1.2s to 350ms.
- Rebuilt async **LLM orchestration** with FastAPI workers, explicit job states, and failure isolation, tripling throughput across 1K+ AI jobs while 5K+ monthly integrations held 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Automated grading for 150+ instructors with **prompt chains** across GPT-4o, Claude, and Qwen-VL plus OCR, handling 10K+ daily requests at 90%+ grading consistency and cutting feedback turnaround 60%.
- Built **model routing** that defaulted to Qwen-VL, sent image answers through GPT-4o Vision OCR, and fell back to Claude Vision on timeouts, logging per-call token cost behind a React dashboard.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Automated cross-platform data-sharing workflows on a microservice architecture, cutting manual data entry 35%, and partnered with Microsoft and GitHub **stakeholders** in Agile sprints from requirement analysis through deployment.
- Led an ETL redesign on AWS Glue and S3 that boosted **pipeline efficiency** 68% and cut storage costs 15%, while API pagination and caching improved response times 30% for low-bandwidth users.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built **resource allocation** and lead management services in TypeScript and Node.js with React and WebSockets, handling real-time assignment tracking and conflict resolution across 2M+ records synchronized between locations.
- Drove **analytics** with Google Analytics on feature usage, load times, and drop-off points, turning insights into product changes that cut resource wastage 42% and raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Translated **client requirements** into an end-to-end booking architecture with 12+ integration points, then built Python APIs for time-slot assignment, pricing, and coupons with Firebase real-time status sync.
- Automated payment webhooks for refunds and disputes, cutting manual handling 70%, and built hourly Databricks **analytics** on peak hours, service demand, and revenue across 50K+ booking records.

Skills

Languages: TypeScript, Python, JavaScript, SQL, Java
Frontend: Next.js, React, TailwindCSS, Shadcn, Dashboards, UI/UX Design
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks
AI: Claude, GPT-4o, Prompt Engineering, RAG, Embeddings, LLM Orchestration, Multi-Model Routing
Data & Infra: PostgreSQL, MongoDB, Supabase, Firebase, Databricks, AWS, Docker, CI/CD
Business & Practices: Twilio, Google Analytics, Resource Allocation, Lead Management, Rapid Prototyping, Stakeholder Management, AI Coding Assistants

Projects

Lead Qualification and Routing Engine | *Next.js, TypeScript, Python, Claude, Twilio, Redis Streams, PostgreSQL*
- Designed a lead-qualification engine where Claude agents with Twilio voice intake score inbound leads and a **capacity-aware router** assigns them to AI or human reps, with **A/B-tested** prompts tracked in Postgres.

HMIS – Hospital Management Information System Live | *TypeScript, MERN, Twilio, Sentry, TailwindCSS*
- Won Smart India Hackathon with a TypeScript operations platform automating appointment booking, rescheduling, and **Twilio** SMS reminders and confirmations across 100+ daily appointments, cutting hospital administrative workload 30%.

PiXMod Live | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB, Stripe*
- Built an AI SaaS platform in **Next.js** with SAM segmentation, Stable Diffusion inpainting, and CLIP search over 100K+ images, adding Stripe subscription flows that lifted premium conversion 18%.